



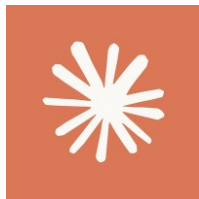
R&D for (Open) Large Language Models in Norway

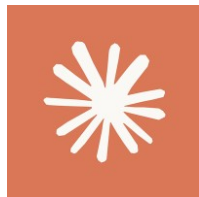
Stephan Oepen, Language Technology Group (LTG), University of Oslo

September 29, 2026 – NorwAI Innovate at Trondheim Tech Festival

0

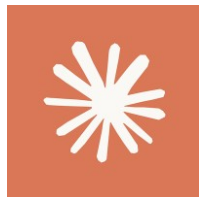
**Background,
Preliminaries**





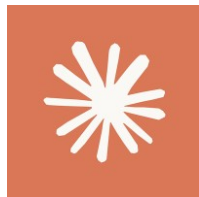
Cultural Norms and Linguistic Identity

- ▶ I helgen var *jeg* på hytten i *skogen* og danset med min *søster*.
- ▶ I helga var *jeg* på hytta i *skauen* og dansa med *søster*a mi.
- ▶ I helga var *eg* på hytta i *skogen* og dansa med *syster*a mi.
- ▶ I helga var *eg* på hytta i *skogen* og dansa med *søster*a mi.



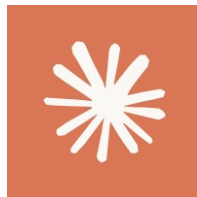
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- ▶ *I helgen* var jeg på hytten i skogen og danset med min søster.
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- ▶ *I helga* var eg på hytta i skogen og dansa med systera mi.
- ▶ *I helga* var eg på hytta i skogen og dansa med søstera mi.



Cultural Norms and Linguistic Identity

- ▶ *I helgen var jeg på hytten i skogen og danset med **min søster**.*
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Cultural Norms and Linguistic Identity

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By Way of Self-Introduction (Gemma-4-21b-it)



By Way of Self-Introduction (Gemma-4-21b-it)



What makes Stephan Oepen at the University of Oslo tick in terms of his academic work?



Reasoning >

Based on his publications, research projects, and academic trajectory, Stephan Oepen's work at the University of Oslo is driven by a desire to bridge the gap between **formal linguistic theory** and **modern machine learning**.

While many in the current NLP (Natural Language Processing) field treat language as a sequence of tokens to be predicted by a "black box" model, Oepen is motivated by the underlying **structure** of language.

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...

Summary: The "Philosophical" Driver

If you had to distill what drives him, it is the belief that **language is not just a statistical pattern, but a structured system**.

“Hallucinations” Remain a Challenge (September 2026)



Here is a breakdown of what "makes him tick" academically:

1. The Quest for Structure (Parsing)

At the core of Oepen's work is **parsing**—the process of analyzing a string of symbols to determine its grammatical structure. He has spent a significant portion of his career exploring:

- **Dependency Parsing:** How words relate to one another in a hierarchy.

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3. Multilingualism and Low-Resource Languages

A significant motivator for Oepen is the democratization of NLP. Much of the field's progress is centered on English. Oepen focuses on:

- **Cross-lingual Transfer:** Using knowledge from a high-resource language (like English) to help a model understand a low-resource language.
- **Universal Dependencies:** He contributes to the effort of creating a consistent framework for grammar across different languages, allowing for a more unified theory of computational linguistics.

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◀ Safari 5G 06:55 98 %

The ChatGPT Lawyer Explains Himself

In a cringe-inducing court hearing, a lawyer who relied on A.I. to craft a motion full of made-up case law said he “did not comprehend” that the chat bot could lead him astray.



1

The Situation in Norway

Februar 5: Renewed and Intensified Collaboration



 Nasjonalbiblioteket

 NTNU



UNIVERSITETET
I OSLO

 sigma2

Språkrådet 

The Norwegian LLM Ecosystem Today (As we See it)



The Norwegian LLM Ecosystem Today (As we See it)



National Library – «Språkmodellfabrikken»

DEVELOPMENT OF APPLICATION-READY MODELS FOR NORWEGIAN AND SÁMI;
NATIONAL BUDGET 2025 & 2026: BASE FUNDING & RIGHTS, NOT RESEARCH.

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Sigma2 – GPU Capacity and HPC Competencies

LARGE-SCALE E-INFRASTRUCTURES AS ESSENTIAL R&D PRECONDITION;
IN COMPARISON TO E.G. FINLAND OR SWITZERLAND, NORWAY IS BEHIND.

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AI Norge & *New Nordics AI* – AI-Driven Innovation, Regulation, Compliance

BROAD VIEW ON AI; NOT FOCUSSED ON RESEARCH OR LANGUAGE MODELS.

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NTNU & UiO (& National Library) – Foundational and Applied Research

SOME ACTIVITY IN NATIONAL AND EUROPEAN RESEARCH PROJECTS;
ISOLATED AND FRAGMENTED; NO AI CENTER HAS LLMS AS ITS FOCUS.



Språkmodellklynge Norge

Om oss



Nasjonalbiblioteket



NTNU



UNIVERSITETET
I OSLO



sigma2 Språkrådet

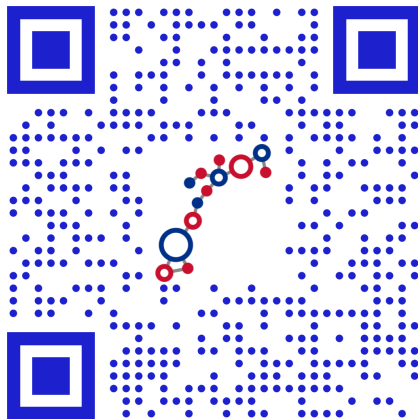
Nasjonal statusmelding: Forskning og utvikling for store, åpne språkmodeller i Norge

kontakt@språkmodellklynge.no

Anbefalinger

Av hensyn til så vel digital som kulturell suverenitet må Norge etablere og drifte et helhetlig nasjonalt økosystem for utvikling, evaluering og anvendelse av egne, trygge og åpne språkmodeller. Det forutsetter blant annet styrket forskningskompetanse og -kapasitet, storskala GPU-infrastruktur, tjenester for data- og modelldeling, samt rammer for samspill mellom grunnleggende forskning og praktisk anvendelse, inkludert styrket nordisk og europeisk samarbeid. Det kreves rask og koordinert handling på regjeringsnivå, inkludert samhandling med for eksempel Norges forskningsråd og Nordisk ministerråd.

Språkmodellklynge Norge anbefaler å:



<https://språkmodellklynge.no/>





1.

Initiate a **fast-moving process towards a holistic national LLM strategy**, with genuine involvement of key Norwegian research environments, together with other national LLM stakeholder, including the Language Council.

Recommendations by the Cluster (1 of 2)



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2.

Create an **AI research center for open language models**, where all actors come together in an academic partnership, including an associated **doctoral training network**. Over five years, **15–20 doctoral degrees** should be conducted.

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1. Initiate a fast-moving process towards a holistic national LLM strategy, with genuine involvement of key Norwegian research environments, together with other national LLM stakeholder, including the Language Council.
2. Create an AI research center for open language models, where all actors come together in an academic partnership, including an associated doctoral training network. Over five years, 15–20 doctoral degrees should be conducted.
3. Strengthen Sigma2 as the national provider of compute capacity, LLM services, and HPC expertise, including increased Norwegian commitment to European initiatives like LUMI-AIF and EuroHPC.



4.

Stimulate Nordic–Baltic research and development collaboration via, for example, data and model sharing, researcher mobility, coordinated training, and an annual summit for researchers and developers of open language models.



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Provide the **legal and practical environment** for data sharing, balancing copyright and privacy with the R&D requirements for **large-scale experimentation** and for **inspection and validation** of models.



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Provide the **legal and practical environment** for data sharing, balancing copyright and privacy with the R&D requirements for **large-scale experimentation** and for **inspection and validation** of models.

6.

Establish Norwegian membership in the **European Alliance for Language Technologies** (ALT-EDIC) and a financial framework for equitable university participation in the **Digital Europe programme**.

Moving on: More Cluster Activity in 2026



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Ongoing: Invitations to Additional R&D and Strategic Partners

SÁMI AI LAB AT SÁMI ALLASKUVLA AND SÁMI GIELLATEKTNO AT UIT;
RELATED ACTIVITIES AT UIB, UIS, NR, SIMULA, . . . ; AI NORWAY.

Moving on: More Cluster Activity in 2026



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Ongoing: Shared Research Infrastructure: Data, Evaluation, Software

UNIFORM SETUP ON OLIVIA OG LUMI; STANDARDIZED EVALUATION;
JOINT WORK WITH ALT-EDIC: NORWEGIAN COMPAR:IA LLM ARENA.

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November 30 & December 1: Nordic–Baltic LLM Builders Summit

40–50 RESEARCHERS FROM DENMARK, FINLAND, ICELAND, SWEDEN, BALTICS;
«HANDS-ON» COORDINATION; E.G. DATA SHARING, TRAINING, EVALUATION.

...

2

**Over There
in Europe**



The Consortium: 21 Partners, 9 European Countries


**Charles
University**

AMD
SILO AI

 alt-edica

EBERHARD KARLS
**UNIVERSITÄT
TÜBINGEN**




e l l i s
INSTITUTE
TÜBINGEN

 **Fraunhofer**
IAIS

 **JÜLICH**
Forschungszentrum

TU/e EINDHOVEN
UNIVERSITY OF
TECHNOLOGY



UNIVERSITY OF HELSINKI



 **UNIVERSITY
OF TURKU**


**ALEPH
ALPHA**

AI
SWEDEN

ellamind

Light

prompsit

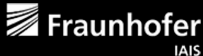
 **BSC**
*Barcelona
Supercomputing
Center*
Centro Nacional de Supercomputación

CINECA


CSC

SURF


The Consortium: 21 Partners, 9 European Countries



UNIVERSITY OF HELSINKI



ellamind

Lighton

prompsit



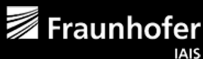
Barcelona
Supercomputing
Center
Centro Nacional de Supercomputación

CINECA



SURF

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UNIVERSITY OF HELSINKI



Barcelona
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Centro Nacional de Supercomputación

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SURF



The Mission: Digital Sovereignty – Transparent AI

Linguistic Equity

36⁺ languages: member states, candidate members, close associates;
emphasize multilingual evaluation: new datasets and benchmarks.

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Open Models & Data

Share models under permissive open-source licenses (e.g. Apache 2.0);
make complete and exact training data generally available for verification.

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Open-source & document full pipeline: code, recipes, results.

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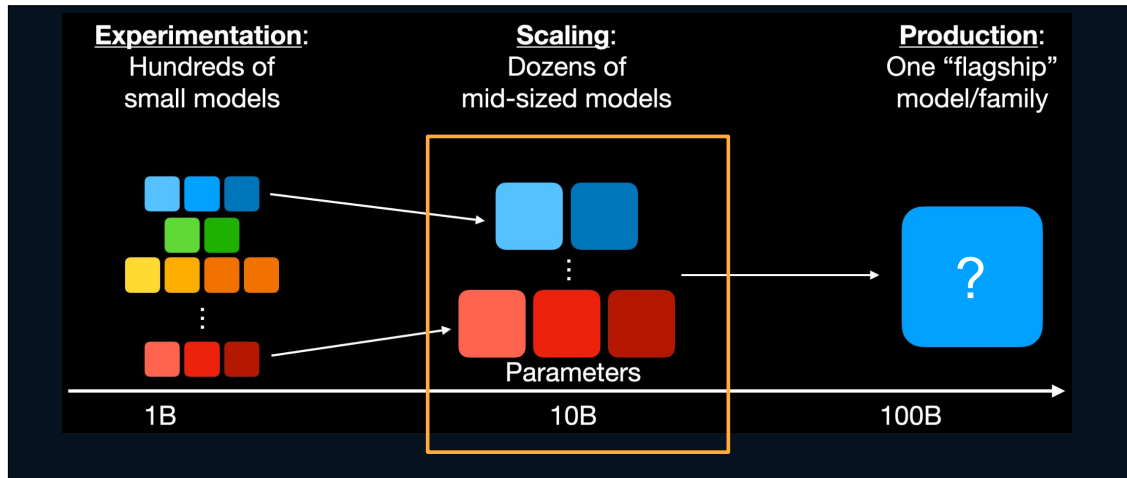
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Reusable Infrastructure

Uniform data management, tools, workflows across EuroHPC ecosystem.

Project Lifecycle: Preparation – Production – Flagship



COLLABORATION – DATA MIXTURES – INFRASTRUCTURE – EVALUATION

EuroHPC Joint Undertaking - Resources for strategic initiatives:

OpenEuroLLM granted substantial **compute** on all major **EuroHPC** supercomputers for developing LLMs for all EU languages



Co-funded by
the European Union

For Example, The OpenEuroLLMs GPU Calculator



	A	B	C	D	E	F	G	H	I	J	K
1											
2		Parameter		Value							
3		Model parameters (N) (billions)		32							
4		Training tokens (D) (trillions)		15							
5		Wasted compute		20%							
6											
7		Training compute (FLOPs)		2.88E+24							
8											
9		Total GPUh by system				Continuous compute days by system and node count					
10		System	TFLOPS	GPUh		Nodes	64	128	256	512	1024
11		JUPITER	400	2.40M		Days on JUPITER	391	195	98	49	24
12		MN5	400	2.40M		Days on MN5	391	195	max nodes 128		
13		LUMI	140	6.86M		Days on LUMI	1116	558	279	140	70
14		Leonardo	150	6.40M		Days on Leonardo	1042	521	260	130	max
15											

NB: estimates do not include time on queue, system downtimes, etc.
Calendar days to complete can be double the continuous compute days or more.

TRAINING FROM SCRATCH MANDATES CROSS-COUNTRY COLLABORATION;
 (OR AN INVESTMENT Á LA SWITZERLAND; OLIVIA IS 1/24 OF ALPS).

3

**Norwegian in
OpenEuroLLM**

2025 Pretraining Data Mix: 10 Trillion Tokens

```
stoepen@uan02:~/training/collection/baby> ./etc/plan.py --budget datamix.txt | egrep "nno|nob"
finepdfs-1.0.0/megatron-lm/nno_Latn: 3,150m tokens of 340m (926.5%).
finepdfs-1.0.0/megatron-lm/nob_Latn: 3,080m tokens of 9,974m (30.9%).
finepdfs-edu-1.0.0/megatron-lm/nno_Latn: 40m tokens of 41m (97.6%).
finepdfs-edu-1.0.0/megatron-lm/nob_Latn: 1,300m tokens of 1,300m (100.0%).
hplt-3.0/megatron-lm/nno_Latn: 9,970m tokens of 1,420m (702.1%).
hplt-3.0/megatron-lm/nob_Latn: 13,700m tokens of 44,588m (30.7%).
nemotron-cc-tower+-0.1/megatron-lm/parallel/tower9b/nno_Latn: 6,780m tokens of 90,132m (7.5%).
nemotron-cc-tower+-0.1/megatron-lm/parallel/tower9b/nob_Latn: 9,320m tokens of 84,161m (11.1%).
```

Norwegian: *Bokmål* and *Nynorsk*

CLOSE TO 50B TOKENS FOR BOTH LANGUAGES: ABOUT 0.5 PERCENT;
ALMOST 40 PERCENT NYNORSK; HEAVILY UPSAMPLED (UP TO TEN TIMES);
USING ALL AVAILABLE NYNORSK, BUT ONLY ABOUT 1/3 OF ALL BOKMÅL;
ONE THIRD OF DATA (17B TOKENS) MACHINE TRANSLATED FROM ENGLISH.

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ENGLISH-FIRST PRETRAINING AS “BEST PRACTICE”: “GENERAL CAPABILITIES”

For Example, Translating from “High-Quality” English



Computer Science > Computation and Language

arXiv:2607.00890 (cs)

[Submitted on 1 Jul 2026]

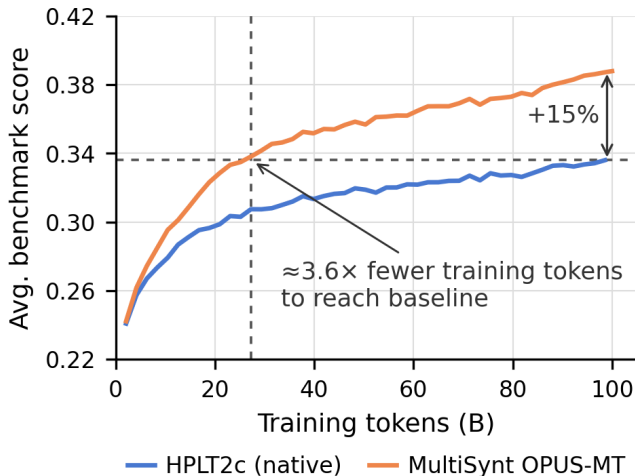
MultiSynt/MT: Trillion-Token Multi-Parallel Pre-Training Data Translated Across 36 Languages

Maximilian Idahl, Jörg Tiedemann, Sampo Pyysalo, David Salinas, Tomasz Galica, Shenbin Qian, Tudor Nicolae Mateiu, Zihao Li, Anna Lokrantz, Fedor Vitiugin, André F. T. Martins, Jenna Kanerva, Filip Ginter, Matthias Lindemann, Tim Isbister, Birger Moell, Jonas Lindh, Jan Hajič, Jenia Jitsev, Andrey Kutuzov, Stephan Oepen, Gema Ramírez-Sánchez

View PDF

HTML (experimental)

Open web-scale pre-training corpora remain concentrated in English, limiting multilingual LLM development. We introduce MultiSynt/MT, an open synthetic parallel corpus with approximately 4.8 trillion target-language tokens across 36 European languages, produced by translating 100 billion high-quality NemoTron-CC tokens with Tower+ and OPUS-MT/HPLT-MT systems. For many medium- and lower-resource European languages, this is the largest openly available pre-training resource. On a broad multilingual benchmark suite, reference LLMs trained on MultiSynt/MT reach the final score of HPLT 2.0, a native-data baseline, using roughly 72% fewer pre-training tokens, and outperform it by approximately 15% relative at a matched 100B-token training budget. Our analyses also identify evaluation blind spots: standard multiple-choice

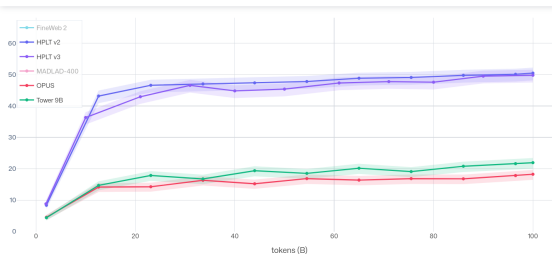


<https://arxiv.org/abs/2607.00890>

Native Benchmarks: Your Mileage May Vary

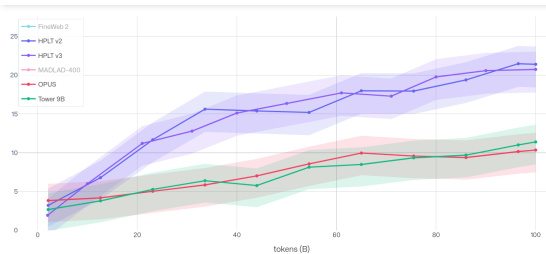
Norwegian – NorIdiom (Bokmål; 0-shot) ▾

Task: NorIdiom (Bokmål) ▾ Shots: 0-shot 5-shot Prompt aggregation: max ▾ Normalization: random baseline ▾
Metric: exact match (first word) (default) ▾ Error bands: shown hidden



Norwegian – NRK Quiz QA (Bokmål; 0-shot) ▾

Task: NRK Quiz QA (Bokmål) ▾ Shots: 0-shot 5-shot Prompt aggregation: max ▾ Normalization: random baseline ▾
Metric: accuracy (default) ▾ Error bands: shown hidden



<https://ltgoslo.github.io/llm-dashboard/multisynt/?lang=Norwegian>

4

**Wrap-Up
Outlook**



4 ESSENTIAL FREEDOMS OF SOFTWARE



0 Run the software whenever you wish, for whatever purpose.



1 Study the source code & make modifications to the software.



2 Give or sell copies of the software.



3 Give or sell copies of your modified versions.

You have the 4 essential freedoms with other useful items that belong to you: clothing, food, simple electrical devices. But most software companies don't want you to have these essential freedoms with software that runs on your various devices, taking away your control over your own devices.

SWITCH TO FREE SOFTWARE!

www.GNU.org

Open Source: Four Essential Freedoms

RUN AS YOU WISH; STUDY AND CHANGE;
REDISTRIBUTE; SHARE DERIVATIVES.

Open-Weights $\neq$ Open Science



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Open Science Language Models

MODEL WEIGHTS (AND CHECKPOINTS);
TRAINING DATA (OR REPRODUCIBLE);
OPEN-SOURCE SOFTWARE STACK;
NON-DISCRIMINATORY TERMS OF USE
PUBLIC INFERENCE ENDPOINTS.

Open-Weights ≠ Open Science



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2 Give or sell copies of the software.



3 Give or sell copies of your modified versions.

You have the 4 essential freedoms with other useful items that belong to you: clothing, food, simple electrical devices. But most software companies don't want you to have these essential freedoms with software that runs on your various devices, taking away your control over your own devices.

SWITCH TO FREE SOFTWARE!

www.GNU.org

Open Source: Four Essential Freedoms

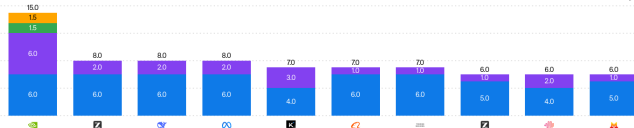
RUN AS YOU WISH; STUDY AND CHANGE;
REDISTRIBUTE; SHARE DERIVATIVES.

Open Science Language Models

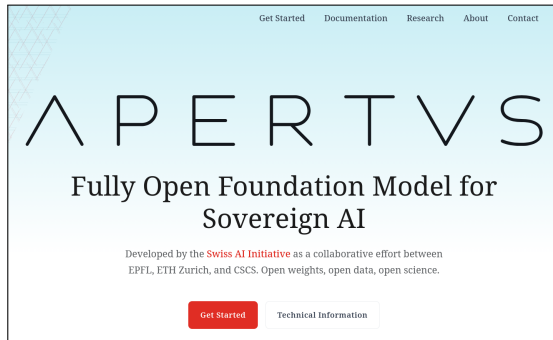
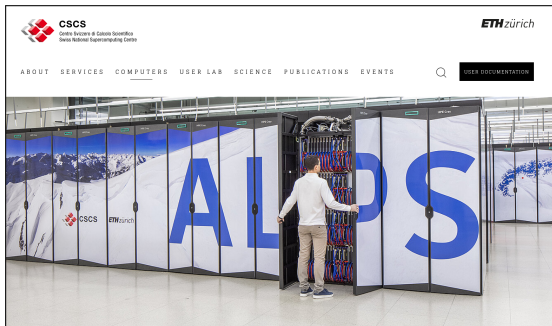
MODEL WEIGHTS (AND CHECKPOINTS);
TRAINING DATA (OR REPRODUCIBLE);
OPEN-SOURCE SOFTWARE STACK;
NON-DISCRIMINATORY TERMS OF USE
PUBLIC INFERENCE ENDPOINTS.

● Transparency - Pre-training Data ● Transparency - Post-training Data ● Transparency - Methodology ● Model Availability

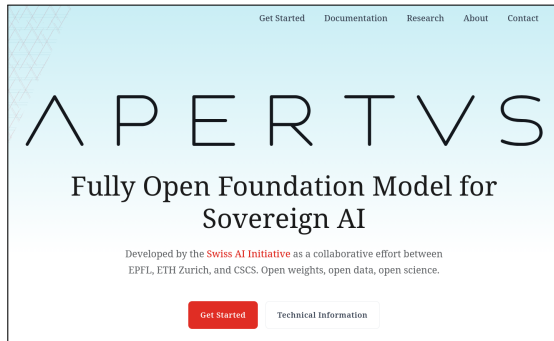
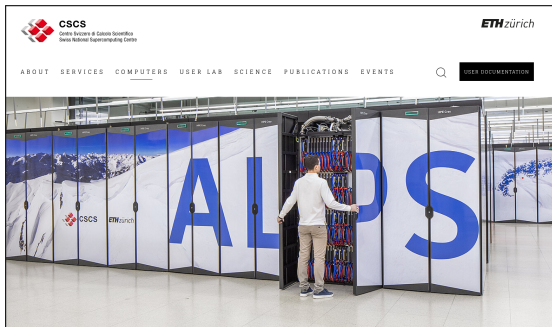
Artificial Analysis



Switzerland, for Example: Alps & Apertus



Switzerland, for Example: Alps & Apertus



THREE SITES (ETH, EPFL, CSCS): COHESIVE TEAM; **PRIORITIZED GPU ACCESS**;
CURRENTLY STRONGEST **FULLY OPEN, MULTILINGUAL** MODELS (9B & 70B).

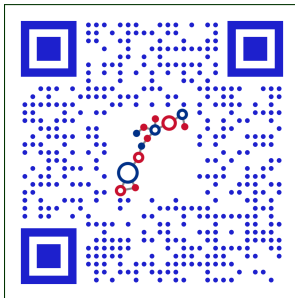


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<https://språkmodellklynge.no/>

<https://www.openeurolm.eu/>